

IoT Based Cardiac Health Monitoring Systems using ECG, PPG and PCG Signals: A Comprehensive Survey

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Abstract— Cardiovascular disorders are the second main leading cause of death globally and comprise 31% share of the total fatalities annually as per WHO. Although under-developing states are more vulnerable to cardiac arrests and heart strokes than developed states the fatality rate in both is burgeoning day by day; even China has declared myocardial infarction fatal because of its destitute infrastructure in traditional healthcare systems where patients fail to seek help in lost consciousness and decease before getting access to any remedy. So, there is a dire need for telehealth monitoring systems to overcome travel distances so that the patients can get quick access to medics in emergencies. In this paper, we have briefly summarized the work of 40 research articles that involve monitoring of cardiac health using Electrocardiography (ECG), Photoplethysmography (PPG), and Phonocardiography (PCG) signals along with the aid of the Internet of Things (IoT). Various components of IoT and combinations of different modules with IoT systems are discussed.

Keywords— *Electrocardiography, Cardiac Health, Internet of Things, Photoplethysmography, Phonocardiography*

I. INTRODUCTION

Healthcare systems involving the Internet of Things (IoT) [1] are raising the standard of human health [2] by transferring the samples [3] acquired by microchips to the IoT cloud [4]. It is a modern digital revolution [5] that is attracting the scientific community all across the world as efficient devices ensure the apex standard of patient care [2] by not only diagnosing anomalies but also prescribing their remedies [6]. This efficient telemonitoring of patients have the following advantages:

- It provides a citadel for data during acquisition and transmission phases [1] and can reduce the chance of late arrivals in severe emergencies [5].
- It enhances productive services by lowering many limitations [8] as it has negligible computational hardware complications [7].

The real-time IoT-based cardiac monitoring device can aid in reducing the fatality rate [8]. As per WHO, the life expectancy of humans will strike 90 years by 2030 [9] because all those hurdles will soon be counteracted by low-cost telemedicine devices [10], which inhibit the Internet of Medical Things (IoMT) to provide quick healthcare access

[11]. Especially, the biomedical field will encounter critical medical disorders to improve the status of human health [12].

IoT with Bluetooth technology can predict pulse, exercise, and oxygen levels [13]; Medical Emergency Response System with IoT aids in better decision-making [14]; Google Speech API with IoT helps blind and deaf patients by transmitting and receiving MP3 audios in real-time [15]; Emotional Computing with IoT can recognize emotions using Electroencephalogram samples [16]; Artificial Intelligence with IoT analyzes bodily mass, temperature, medical reports [17] & pulse by WSN [18]; Physical Activity Recognition & Monitoring (PARM) Model with IoT can analyze pressure position [19]; Deep Brain Stimulation can access all alterations generated in metabolic activities, expressions and movements of the brain with IoT [20]; Computer-Human Interaction [21] and Synergy-based bionic limbs [22] can be manufactured using EMG signals with IoT.

II. MAJOR COMPONENTS OF AN IoT SYSTEM

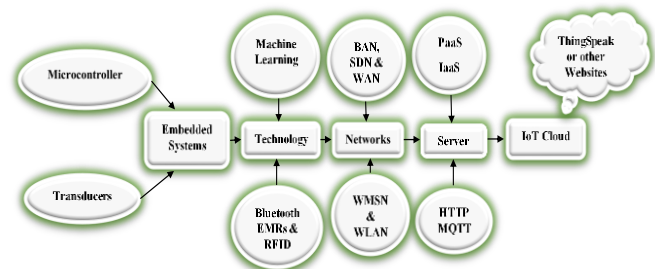


Fig 1: Block Diagram for major components of IoT based System

IoT is a combination of transducers, wireless communication systems, computing, internet protocol, and much more [23]. The major components of an IoT-based healthcare system are illustrated in Fig. 1.

A. Embedded Systems

These systems are mainly composed of wearable gadgets, microcontrollers, and sensors. Table 1 summarizes the details of embedded systems that involve IoT.

TABLE 1: DETAILS OF EMBEDDED SYSTEMS HAVING IOT

Ref	Models	Functions
[24]	LCHM and CoSHE	Low-Cost Health Monitoring Model & Cloud-based Smart Home Environment can acquire data using transducer nodes & robotic assistants.
[25]	MCOs	Myriad of Medical Connected Objects can monitor patients effectively with the aid of Ontology Domain (HealthIoT-O).
[5]	BARC	Bluetooth in mobiles can operate Tele-ECG devices and transmit samples to the cloud via MMS.
[24]	Environmental Indicators	These can access health conditions along with raising alarms to inhibit potential.

B. Technology

The IoT-based systems involve various technologies for the storage and display of data along with performing real-time analysis remotely using Machine Learning techniques [26]. Some of them are Bluetooth Technology[4], Radio Frequency Identification Technology [5], Electronic Medical Records [27], and GSM, GPRS, ADSL & GUI Modules[26].

C. Networks

The combination of Body Sensor Networks (BSNs) with Internet-of-Things (IoT) aids in the creation of smart and remote healthcare systems [28]. Table 2 presents different combinations of networks with IoT.

TABLE 2: DIFFERENT NETWORKS OF IOT

Ref	Network	Functions
[5]	BAN	Wired and Wireless Network Contact using Body Area Network can analyze the link between cloud information & wearable devices.
[7]	WBAN	WBAN based Healthcare Environment handles telemonitoring of MEGG and generates an ample quantity of sensory data.
[29]	SDN	Software-Defined Network collects data with smartphones & displays predicted health status.
[26]	WMSN & WLAN	Wireless Medical Sensor Network provides novel and smart solutions while Wireless Local Area Network transmits data in real-time.
[30]	SN	An amalgam of IoT with Social Networks results in Social IoT (SIoT) which can aid in maintaining social relations in a better way.

D. Server

Compressed Sensing (CS) is the simplest encoding tack for decimating complexities at the transmitter portion and for easing the measurement progress in WBAN [27]. Servers like High-performance Microcontroller Unit (MCU) aids in signal processing applications [28]; HTTP and MQTT protocols deliver data timely; The cloud is accessible by Wide Area Network using PaaS (Platform as a Service) and IaaS

(Infrastructure as a Service) [5]; Mobile Apps and Webpages are helpful in data management and visualization.

III. IOT BASED CARDIAC HEALTH MONITORING SYSTEM

A. ECG Based IoT Systems

In 2016, a wearable monitoring node, comprising HTTP and MQTT protocols, collected ECG data & transferred it to the IoT cloud for the initial determination of cardiovascular disorders [24]. Then, Wireless Android Smartphones were also used in this regard, and synchronization was carried out using Rest API in IoT Cloud [32]. Later on, Zigbee was introduced which aided the patients to get in touch with medics at any time [33]. The quick access to medics was also achieved through Raspberry Pi-based health monitoring device using GSM Technology and IoT server [34] followed by usage of nano-material enabled dry ECG electrodes for the effective data acquisition instead of conventional [35].

Temperature data and ECG were acquired by smartphone. After analyzing data during myocardial infarction, it was sent to the cloud using a Low Energy Bluetooth communication system [36]. In 2018, Health-Fog was introduced which analyzed cardiac ailments and provided a citadel during the transmission phase. Fog-Bus ensured the accuracy, bandwidth, latency, and time taken for execution. Data was taken from Physio-Bank for the system named as ECG-based CA mechanisms for calculating the results for the known, unknown and blind-modeled attackers [37]. Later on, Holter systems were used as their sampling frequency was the same as that of ECG signal, i.e., 128Hz [38].

Cloud-based HRV detection system acquired and transmitted data wirelessly by using Serial Peripheral Interface (SPI) [39] and in remote areas, real-time ECG monitoring using IoT was carried out by a portable system comprising 3 leads [40]. In 2019, CS-SCGN was introduced for feature extraction after analysis and transmittance phases which were carried out by DBBD based CS encoding for achieving a plug and play architecture [41]. Later on, bodily motions and ECG patterns were monitored by wearable low power communication embedded system-based smartphones [42]. Then, burgeoning and diminishing cardiac ailments were classified by Machine Learning [43].

In 2020, Body Sensor Networks were used with the Data-Analytic tacks and IoT for manufacturing a device that is efficient, remote, and wireless albeit expensive with irrational safety levels [28]. An intelligence system named as Physical Activity Recognition and Monitoring (PARM) Model was proposed for monitoring temperature, pressure position, and heart rate along with counteracting challenges in the domain of IoT and providing data with safety [19].

TABLE 3: SUMMARY OF ECG BASED IOT SYSTEMS FOR CARDIAC HEALTH MONITORING

Ref	Year	Device	Technology	Dataset and Bio-Signal	Comments
[24]	2016	Zigbee and AD8232 ECG Sensor	Bluetooth & Wi-Fi	HTTP and MQTT databases of ECG	Delivered pictorial presentation & lowered tardy arrivals using Wi-Fi
[32]	2016	Android Smartphones	Wavelet Transform (WT)	ECG Samples	98% accuracy
[33]	2016	Zigbee	Signal processing	ECG Samples	Satisfactory outcomes
[34]	2017	Raspberry Pi	GSM Module	Cellular Network of ECG and Temperature Samples	Samples were sent by IoT server with minimal data loss
[35]	2017	M2M & CNT electrode	Local Area Network (LAN)	ECG Samples	Better results were obtained

[36]	2017	Arduino uno	Low Energy (LE) Bluetooth	ECG & Temperature Samples	Good Results
[37]	2018	Walsh-Hadamard Transform (WT) CA Mechanism	Health-Fog, Fog-Bus, and MODWT for distinguishing alterations of R waves	MIT-BIH databases of ECG	Results were 99.5% for Blind, 93.5% for known, and 91.8% for unknown
[38]	2018	Holter Devices	MODWT, DWT, and HWT	MIT-BIH and Arrhythmia Databases of ECG	94-96% with MODWT, 85-87% with DWT & 89-93% with HWT
[39]	2018	ECG Sensor AD8232, MSP430 Microcontroller & CC3100 Wi-Fi	PubNub and Serial Peripheral Interface (SPI)	Heart Rate Variability (HRV) databases of ECG	TCP/IP stack streams data to PubNub IoT Web Interface with minimum complications
[40]	2018	TIVA Microcontroller	CC3100 Wi-Fi Booster Pack	ECG Samples	Effective results were obtained
[41]	2019	Atmega32 Microcontroller	Fog Detection	Sensor Node of ECG	CS-SCGN was possible
[42]	2019	Smartphone	GPS sensors & LP Bluetooth	ECG and Body Acceleration	Both were monitored successfully
[43]	2019	Machine Learning	KNN & SVM	ECG Samples	88.52% accuracy was achieved
[28]	2020	Body Sensor Networks (BSNs)	SVM, CWT, DWT and Selective Encryption	ECG Samples of Emergency Medical Response System	Overall 70% and 81%, 71% & 75% for Db2, Db4 and Db8
[19]	2020	Wearable Sensor System	WLAN & Wi-Fi	ECG, EEG, SpO2 & Temperature	98% accuracy

B. PPG Based IoT Systems

Bluetooth technology was initially used to transmit the heart rates of a fetus from the fifth to the last week of pregnancy to avoid severe complications at delivery [44]. Within the same year, the connectivity with wearable gadgets and smartphones was made possible when deep learning was applied to IoT after acquiring samples with accelerometer transducers [45]. Then, LoRaWAN IoT Eco-System Network was applied to transmit data towards cloud servers [46].

The conflation of Mobile Edge Computing (MEC) with Software Defined Network (SDN) resulted in a Novel e-health Service Platform where voice and gesture were controlled by Real-time Heart Attack Mobile Detection Service (RHAMDS) [47]. Then, wearable gadgets were introduced for reminding the patients about their daily medications. Whenever their heart rate increases or decreases beyond the set limit, an alarm was raised and an email was also sent immediately [48].

Machine Learning was also proposed for raising alarms during risky situations or when patients endured acute stress [49]. Also, a prototype design involving GUI was introduced which monitored heartbeat using smartwatch Samsung gear S3 [64]. Later on, a Fuzzy interface was introduced after

describing the usage of IoT applications in the health sector along with required crucial units [50]. In 2018, wearable gadgets were introduced for seven months of examination of heartbeats, and the highest results were achieved using Random Forests with 10-fold cross-validation [51]. Gradually, with the Global System of Mobile Communications (GSM), an alert and mathematical detection system were also designed within the same year [52].

In 2019, 30% testing and 70% training were performed on 1500 subjects' data in higher-order Boltzmann deep belief neural network (HOBDBNN) based on the IoT system recognized heart disorders [53]. The amalgam of Photoplethysmography (PPG) and Convolutional Neural Networks led towards the better acquisition of signals and yielded better results by categorizing peculiarities among reliable and unreliable [24].

In 2020, Wireless and Wearable Intelligent Systems diagnosed physical ailments more accurately in real-time [13] followed by enhancement of MPM performance using Support Vector Machine algorithm [54]. Finally, signals were distinguished using Deep Learning Neural Network involving authentication, encryption & classification by substitution cipher (SC), PDH-AES, and DLMNN classifier [55].

TABLE 4: SUMMARY OF PPG BASED IOT SYSTEMS FOR CARDIAC HEALTH MONITORING

Ref	Year	Device	Technology	Dataset and Bio-Signal	Comments
[44]	2016	Smartphone	Bluetooth Technology	Fetus heart rate samples	Satisfactory Results
[45]	2016	Mobile phones & Wearable devices	Accelerometer transducers	TROIKA dataset of PPG	10-fold cross validation error margin of 4.88% was left
[46]	2017	LoRaWAN IoT	LOMB HRV Analysis	Heart Rate Variability dataset	50%-30% chances of error
[47]	2017	RHAMDS Watches	SDN Wireless Node & Cloud Server	PPG Samples	Satisfactory results
[48]	2017	Wearable device	Email	Heart-rate Samples	Beneficial for patients
[49]	2017	Wearable Gadget	Machine Learning (ML)	Heart-beat Samples	Results were 66% (Logistic Regression) and 68% (SVM)
[56]	2017	Wearable Smartwatch Samsung Gear S3	Signal processing and GUI code	Heart-rate Samples	Libraries Nodejs, JavaScript, and WebSocket were involved
[50]	2018	M2M	Fuzzy Interface	Constrained Application Protocol	Satisfactory Results
[51]	2018	k-means algorithm	Cloud server	PPG Samples	97.9% accurate results
[52]	2018	Arduino UNO	Fuzzy c-means clustering	PPG Samples	82% accuracy
[53]	2019	MATLAB Simulation	Boltzman Order Classification Approach	Abnormal Heart Signals of HOBDBNN, GA-TRFNN, SCNN-SVM and PFFBPNN	99.5% (HOBDBNN), 95.6% (GA-TRFNN), 96.2% (SCNN-SVM) & 98.5% (PFFBPNN)
[24]	2019	E4 and PulseOn Wearable Devices	Convolutional Neural Networks Approach and Deep Learning	PPG Samples	This method was 85% reliable and 83% unreliable
[13]	2020	Health-Monitoring Wristband	Bluetooth, Wireless and Intelligent Health Service Terminal System	PPG, Blood Pressure, Blood Oxygen & Pulse	Satisfactory outcomes were achieved
[54]	2020	Machine Learning Devices	Support Vector Machine (SVM) Algorithm	Heart Rate, Respiration, Oxygen Saturation & Temperature	95% outcomes

[55]	2020	Wearable IoT Sensor	Deep Learning Modified Neural Network (DLMNN) Classifier	Diabetes, Cancer and Chronic Respiratory Maladies	Results were 95.87% accurate
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C. PCG Based IoT Systems

Internet of Medical Things (IMedT) was initially used to monitor cardiac activity using PCG signals [57] but later on, Noise-Filtration techniques were applied to distinguish anomalies more accurately and this method yielded 40% more efficient outcomes than the former because of the reduction of artifacts from heart sounds [58].

Gradually, scientists worked on the amplification of the sample and its conversion into a digital signal before sending it to medics through the IoT cloud [59]. Recently, when Machine Learning tasks are involved in this whole process, the differential privacy, as well as protection of data, has been increased to a greater extent [60]. Finally, the cardiac sounds of patients are classified into normal, mild, and critical using different processing techniques [61].

TABLE 5: SUMMARY OF PCG BASED IOT SYSTEMS FOR CARDIAC HEALTH MONITORING

Ref	Year	Device	Technology	Dataset	Comments
[57]	2016	5G mobile networks	IMedT involving manager & agent	PCG Samples	Samples were analyzed
[58]	2017	Phonocardiogram	Ensembling Learning Model	MIT-Physionet dataset	85% accuracy
[59]	2018	Acoustic Stethoscope	IoT Cloud	Heart Sounds	More reliable method
[60]	2019	Automated Remote System	Robust Machine Learning Tasks	MIT-Physionet datasets	85% accuracy
[61]	2019	IoT-based Intelligent Decision-Making System	Edge Processing and Neural Networks	Real-time dataset of PCG	Patients were categorized

D. Data-Fusion Based IoT Systems

Initially, real-time data was obtained and transmitted for the early diagnosis and timely rehabilitation using Mobile health devices in which Health Insurance Portability and Accountability Act acts as a citadel for data [62]. Then, the DWT technique was added up to reduce BW, noise, and artifacts during the data acquisition phase, especially for ECG samples [4]. In 2019, a smart & novel Fog-based algorithm comprising an IoT system with a back-end server was

introduced for monitoring blood glucose and cardiac parameters remotely in real-time [7].

Later on, for better online assessment, inexpensive IoT devices involving MIT-APP Inventor 2 and iOS versions were taken into consideration [31]. Also, an inexpensive wearable gadget was introduced whose security level was reliable to such extent that only authorized personnel can get access and whenever any malformation is diagnosed, an email is ultimately forwarded for examination [2].

TABLE 6: SUMMARY OF DATA-FUSION BASED IOT SYSTEMS FOR CARDIAC HEALTH MONITORING

Ref	Year	Device	Technology	Dataset	Comments
[62]	2016	M-Health Device and Zigbee	Health Insurance Portability and Accountability Act	ECG, PPG, and Blood Sugar Samples	IoT servers transmitted data with minimal loss
[4]	2017	Smartphones, Personal Digital Assistant (PDA) laptops	Bluetooth, Cellular Technologies (GSM and GPRS), and broadband Wired Technologies (ADSL)	ECG, Heart rate, Blood pressure, SpO2, Pulse, Blood fats & glucose Samples	Porous graphene (PG) foam was manufactured by deducing artifacts
[7]	2019	WBAN IoT Node	Telemonitoring system applying Fog between clouds & gateways	MITDB & CPM Databases of ECG, PPG, EMG, and EEG	Enhancement of sensor nodes lead towards better outcomes
[31]	2019	CPU, GPU, LCHM & SFG	MIT APP Inventor 2 and iOS version	ECG & BPS Samples of RNN, LCHM, FogCepCare, and ECGH	The online assessment proved to be very valuable
[2]	2020	Wireless & Mobile Heart Monitoring Systems	Wireless Sensor Technologies & Body Area Network (BAN)	ECG & PPG Samples of IEEE Xplore and Springer	This system proved to be very reliable in terms of the privacy

IV. RESULTS AND DISCUSSION

The maximum accuracy achieved by ECG is 98% using Wearable Sensor System with the help of WLAN & Wi-Fi using SVM & KNN Classification. For PPG, the highest results are 95.87% using Wearable IoT Sensor with Deep Learning Modified Neural Network Classifier. For PCG, 85% accuracy is achieved using an Automated and Remote System and infused data, the samples were transmitted through IoT Cloud with minimal loss.

Although ECG has the highest accuracy, PPG is more reliable as it is the least vulnerable to data loss because it neither has any complex analog waveform nor any anatomical noises endured during the data acquisition process. ECG has a complex QRS interval which is most vulnerable

to data loss during the transmittance phase. Also, internal anatomical noises like lungs sounds can alter the desired outcomes. On the other hand, PCG has less accuracy due to environmental artifacts.

V. CONCLUSION AND FUTURE WORK

These IoT-based devices involving ECG, PPG, and PCG signals help monitor cardiac activity. Especially, in present times, Covid-19 can also be detected using these telemedicine technologies. It can also be applied to a pulmonary system to make their diagnosis much more reliable as these disorders are mostly contagious so their telemonitoring will be very valuable in this regard.

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